Davies-Bouldin Index (DBI) Explained

The **Davies-Bouldin Index (DBI)** is a clustering evaluation metric that measures:

1. **How compact each cluster is** (intra-cluster similarity)
2. **How well-separated clusters are** (inter-cluster distance)

Unlike the Silhouette Score:

- **Higher Silhouette Score = Better**
- **Lower Davies-Bouldin Index = Better**

Intuition

A good clustering solution should have:

✅ Points within a cluster close together

✅ Different clusters far apart

Bad clustering:

❌ Large spread within clusters

❌ Clusters very close to each other

DBI quantifies this.

Formula

For two clusters i and j:

$$R_{ij} = \frac{S_i + S_j}{M_{ij}}$$

Where:

- S_i = average distance of points in cluster i from its centroid (cluster spread)
- S_j = average distance of points in cluster j from its centroid
- M_{ij} = distance between centroids of clusters i and j

The Davies-Bouldin Index is:

$$DBI = \frac{1}{K}\sum_{i=1}^{K} \max_{j\neq i}\left(\frac{S_i + S_j}{M_{ij}}\right)$$

Where:

- K= number of clusters

Simple Example

Suppose we have two clusters:

Cluster A

```
1, 2, 3
```

Centroid:

$$C_A = \frac{1+2+3}{3} = 2$$

Cluster B

```
10, 11, 12
```

Centroid:

$$C_B = \frac{10+11+12}{3} = 11$$

Visualization:

```
1    2    3                    10    11    12
|---Cluster A---|              |---Cluster B---|
       C_A                            C_B
```

Step 1: Calculate Cluster Spread

Cluster A

Distance from centroid 2:

```
|1-2| = 1
|2-2| = 0
|3-2| = 1
```

Average:

$$S_A = \frac{1+0+1}{3}$$
$$S_A = 0.67$$

Cluster B

Distance from centroid 11:

```
|10-11| = 1
|11-11| = 0
|12-11| = 1
```

Average:

$$S_B = 0.67$$

Step 2: Calculate Distance Between Centroids

$$M_{AB} = |\ 11 - 2\ |$$
$$M_{AB} = 9$$

Step 3: Calculate Similarity Ratio

$$R_{AB} = \frac{0.67 + 0.67}{9}$$
$$R_{AB} = 0.149$$

Since there are only two clusters:

$$DBI = 0.149$$

Interpretation

```
DBI = 0.149
```

Very low value → Excellent clustering.

Why?

- Small cluster spread
- Large centroid separation

Bad Clustering Example

Suppose:

Cluster A

```
1,2,8
```

Cluster B

```
9,10,11
```

Visualization:

```
1   2      8  9   10  11
|----A----| |----B----|
```

Clusters overlap.

Centroids:

$$C_A = 3.67$$
$$C_B = 10$$

Spreads become larger and centroid distance becomes smaller.

Result:

```
DBI ≈ 1.2
```

Higher DBI indicates poorer clustering.

Typical DBI Values

DBI	Interpretation
< 0.5	Excellent
0.5 – 1.0	Good
1.0 – 2.0	Acceptable
> 2.0	Poor
Very High	Significant overlap

There is no strict universal threshold; DBI is most useful for comparing clustering solutions.

Choosing K Using DBI

Suppose:

K	DBI
2	1.25
3	0.82
4	0.48
5	0.62
6	0.79

Graph:

```
DBI
1.4 | *
1.2 |
1.0 |
0.8 |      *
0.6 |            *
0.4 |                *
0.2 |
    +-------------------
      2  3  4  5  6
            K
```

Lowest value:

```
K = 4
```

Therefore, 4 clusters are preferred.

Comparison with Silhouette Score

Metric	Goal
Silhouette Score	Maximize
Davies-Bouldin Index	Minimize
Calinski-Harabasz Index	Maximize

Example:

K	Silhouette	DBI
2	0.62	0.92
3	0.75	0.61
4	0.83	0.42
5	0.78	0.58

Both metrics suggest:

```
K = 4
```

Real-World Example: Customer Segmentation

A retailer clusters customers using:

- Income
- Spending Score

Results:

K	DBI
2	1.10
3	0.79
4	0.44
5	0.67

Since **K = 4** has the lowest DBI, the customer groups are most compact and best separated.

Interview Answer

The Davies-Bouldin Index is an internal clustering evaluation metric that measures the average similarity between each cluster and its most similar neighboring cluster. It considers both cluster compactness and separation. Lower values indicate better clustering because clusters are tightly packed and far apart. It is commonly used to compare different values of K in K-Means and select the clustering configuration with the lowest DBI.

Memory Trick

```
Silhouette  ↑ Better
DBI         ↓ Better
```

- **High Silhouette = Good**
- **Low Davies-Bouldin = Good**
- **High Davies-Bouldin = Overlapping clusters**